MA 5104: Statistical Inference and Learning

Module 4: Learning

Lectures 34: Principal Component Analysis

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Outline

1. The Problem
2. Finding the Direction
3. Using It
4. Bringing in y
5. Closing

You Cannot Look at Twenty Columns

To understand a dataset we look at it. With p variables, the number of pairwise scatterplots is

$$\frac{p(p-1)}{2}$$

p	plots
5	10
10	45
20	190
200	19,900

And even all 190 plots would not show you a pattern that lives in three variables at once.

We want a handful of new variables that carry most of what the original ones carried.

Two Columns, One Direction



Two measurements on 180 items. How many numbers do you really need to say where a point is?

Formally two. But the cloud is long in one direction and thin in the other: tell me how far along the long axis a point sits and you have said nearly everything.

The second direction carries very little (mostly measurement noise).

The data have two columns but one direction. Everything that follows is machinery for finding that direction when $p = 200$ and you cannot see the picture.

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What Is a "Direction"?

A direction in $\mathbb{R}^p$ is a vector ϕ of unit length, $\|\phi\| = 1$.

To summarise an observation x by that direction, record how far along it the point sits:

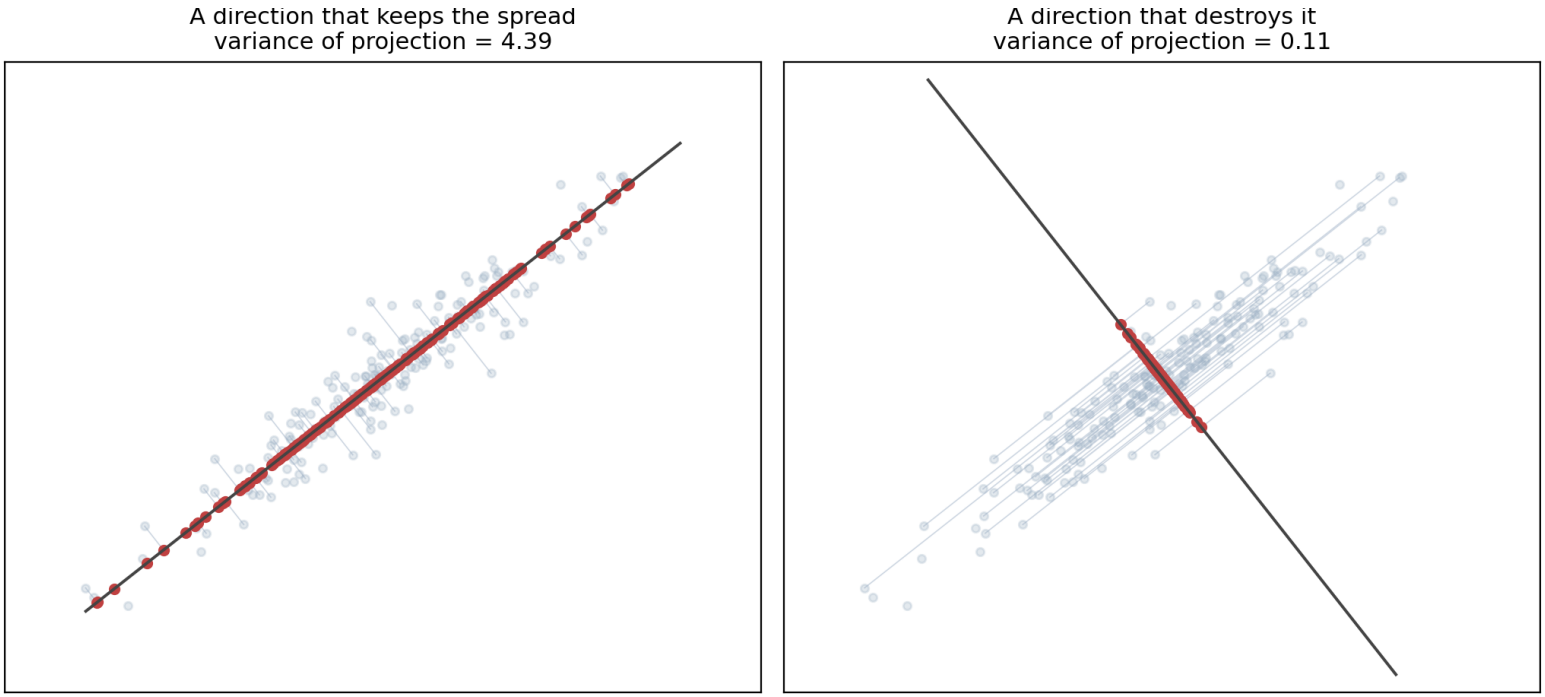
$$z = \phi^\top x = \phi_1 x_1 + \phi_2 x_2 + \cdots + \phi_p x_p$$

One number instead of p. This is a projection: p columns in, one column out.

- The ϕ_j are the weights: how much each original variable contributes.
- $\|\phi\| = 1$ stops us cheating — without it we could inflate the spread of z by simply scaling ϕ up.

Centre first. Subtract each variable's mean, so the cloud sits at the origin and z measures displacement from the centre.

Which Direction Should We Keep?



On the right, points that were far apart land on top of one another. Information about who differs from whom has been destroyed.

Spread is not an aesthetic preference. It is the only thing that distinguishes one observation from another: a direction with no spread gives every point the same number, and so tells you nothing.

Writing the Criterion Down

Let $X \in \mathbb{R}^{n \times p}$ be the *centred* data, rows $x_1, \ldots, x_n$. The sample covariance matrix is

$$S \;=\; \frac{1}{n-1} X^\top X \;\in\; \mathbb{R}^{p \times p}, \qquad S_{jk} = \widehat{\mathrm{Cov}}(x_j, x_k)$$

The projected values are $z_i = \phi^\top x_i$, and their variance is

$$\mathrm{Var}(\phi^\top x) \;=\; \phi^\top S \, \phi$$

So the question "which direction keeps the most spread?" becomes

$$\boxed{\phi_1 \;=\; \arg\max_{\|\phi\|=1} \; \phi^\top S \phi}$$

A clean optimisation problem. The covariance matrix entered because we needed it, not because we were doing notation.

Solving It

Maximise $\phi^\top S\phi$ subject to $\phi^\top\phi = 1$. Lagrange:

$$L(\phi, \lambda) = \phi^\top S\phi - \lambda\left(\phi^\top\phi - 1\right)$$

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So ϕ must be an eigenvector of S. But which one? Multiply on the left by $\phi^\top$:

$$\phi^\top S\phi = \lambda\,\phi^\top\phi = \lambda$$

The eigenvalue *is* the variance we were maximising. So take the eigenvector with the *largest* eigenvalue. That is the first principal component direction, and λ_1 is the variance along it.

The Second Direction, and the Rest

Having used the best direction, ask for the next one — subject to being orthogonal to the first, so that it carries new information rather than repeating what we have.

The answer is the eigenvector with the second largest eigenvalue, and so on. Writing the eigendecomposition

$$S = P\Lambda P^\top, \qquad P = [\phi_1, \ldots, \phi_p], \qquad \Lambda = \mathrm{diag}(\lambda_1 \geq \cdots \geq \lambda_p \geq 0)$$

gives all p directions at once.

- S is symmetric, so the eigenvectors are automatically orthogonal — the constraint comes for free.
- S is positive semi-definite, so every $\lambda_j \geq 0$. They are variances; they could not be negative.
- The new variables $Z = XP$ are uncorrelated with each other. We have rotated into a coordinate system where the columns no longer duplicate one another.

Three Words You Will See in Any Output

- **Loadings** the entries of ϕ_k — how much each original variable contributes to component k. A property of the *variables*.
- **Component** the new variable itself, $\mathrm{PC}_k = X\phi_k$.
- **Scores** the values $z_{ik} = x_i^\top \phi_k$ — where observation i sits along component k. A property of the *observations*.

Worked example. Two variables, marks in Maths and in Physics. Suppose $\phi_1 = (0.71,\ 0.71)^\top$.

Then $\mathrm{PC}_1 = 0.71\, x_{\text{maths}} + 0.71\, x_{\text{physics}}$ — the loadings are both 0.71, so this component is essentially "overall performance".

A student whose centred marks are $(10, 12)$ has score $z_1 = 0.71(10) + 0.71(12) = 15.6$.

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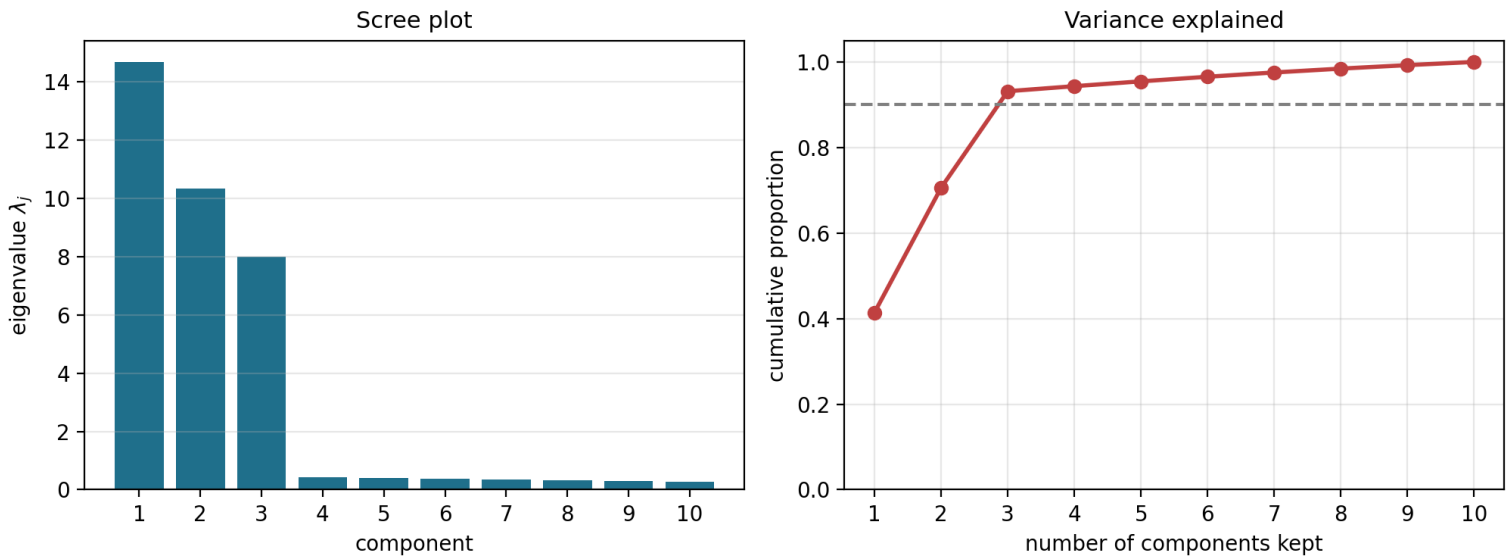
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How Many Components Do We Keep?

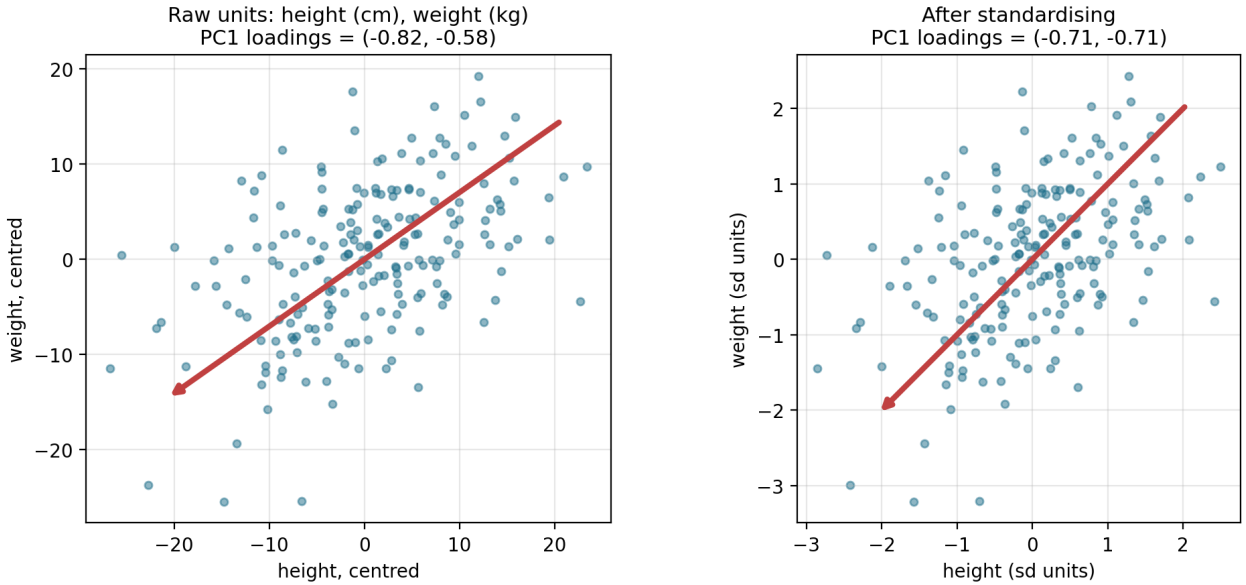
Total variance in the data is $\mathrm{tr}(S) = \sum_j \lambda_j$, so the proportion captured by the first k components is

$$\frac{\lambda_1 + \cdots + \lambda_k}{\lambda_1 + \cdots + \lambda_p}$$



Here three components carry 93% of the variance of ten variables. Look for the elbow, or fix a target such as 90% — but note that neither rule is a theorem, and the choice is a judgement.

Scaling Is Not Optional



Same people, same two measurements. On the left, height in centimetres has far more numerical variance than weight in kilograms, so PC1 is almost entirely height. Standardise, and both variables get a say.

PCA maximises variance, and variance has units. Record height in millimetres and its variance multiplies by 100 — the first component changes, though nothing about the people did. Standardise unless the variables are genuinely in the same units.

The Same Thing, Computed Better

In practice nobody forms S and diagonalises it. Take the singular value decomposition of the centred data directly:

$$X = UDV^\top$$

Then

$$S = \frac{1}{n-1} X^\top X = \frac{1}{n-1} V D^2 V^\top \qquad \Longrightarrow \qquad \phi_k = v_k, \qquad \lambda_k = \frac{d_k^2}{n-1}$$

- Identical answer, better numerics: forming $X^\top X$ squares the condition number.
- The d_j appeared in the ridge shrinkage factor $d_j^2/(d_j^2+\lambda)$. Ridge shrinks hardest along the directions PCA discards first.
- Keeping the top k components gives the *best possible* rank-k approximation to X in squared error.

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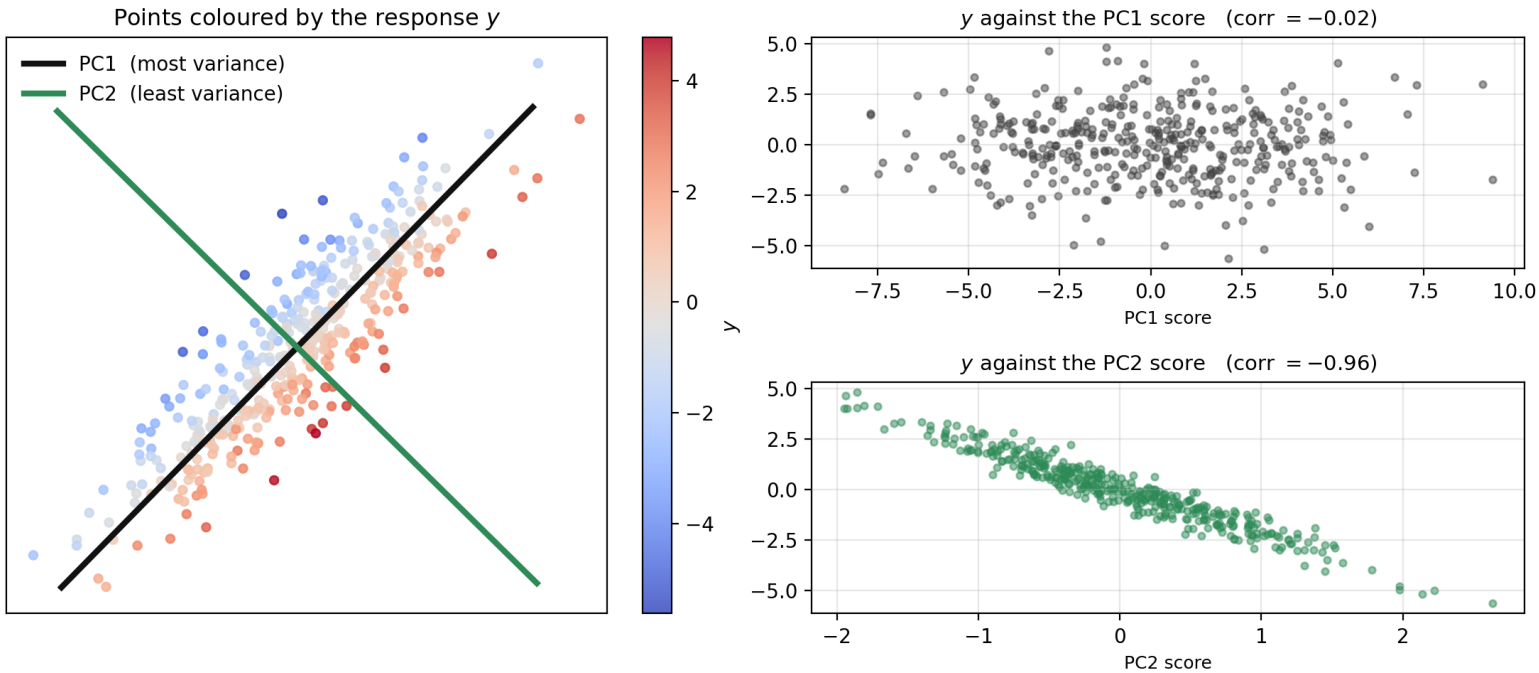
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PCA Never Looks at the Response

Everything so far used X alone. That is a feature when exploring, and a hazard when predicting.



PC1 carries 94% of the variance in X and is useless for y (correlation -0.02). PC2 carries 6% and explains y almost perfectly (correlation -0.96).

Principal Components Regression

PCR. Compute the first k principal components of X, then regress y on those k new variables instead of the original p.

- Fewer predictors, and they are uncorrelated, so the fit is stable even when the original columns were badly collinear.
- k is chosen by cross-validation, exactly as λ was.
- Often works well, because in real data the directions with large variation frequently *are* the ones that drive the response.

But the previous slide is a real possibility

PCA ranked PC1 first and discarded PC2 — correctly, by its own criterion. It never saw y, so it could not know which direction mattered. If we want the components to be useful for prediction, the response has to get a vote.

PLS: Let the Response Choose the Weights

We want one new variable $z = \phi_1 x_1 + \cdots + \phi_p x_p$ that is useful for predicting y. How should we choose the weights?

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Naturally: give more weight to the predictors that move with y. Standardise the columns, then set

$$\phi_j \;\propto\; \widehat{\mathrm{Cov}}(x_j,\, y) \qquad \text{(equivalently, the correlation between } x_j \text{ and } y\text{)}$$

and rescale so $\|\phi\| = 1$. That single rule *is* the first direction of partial least squares.

Compactly, it is the solution of

$$\max_{\|\phi\|=1} \; \widehat{\mathrm{Cov}}\big(X\phi,\; y\big)^2$$

the same shape of problem as PCA, with Var replaced by Cov with y.

PLS: The Rest, and the Verdict

Later directions. Remove from each column of X the part already explained by z_1, then apply the same weighting rule to what is left. This keeps the new directions from repeating one another, exactly as orthogonality did in PCA. Stop after k directions; choose k by cross-validation.

- PCA asks: in which direction do the predictors vary most?
- PLS asks: in which direction do the predictors vary *in a way that tracks y*?

P

LS cannot be fooled the way PCR was on the earlier slide. But in practice it often does little better than PCR, and sometimes worse: chasing y means also chasing the noise in y. Lower bias, higher variance — the same trade as everywhere else in this course.

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Two Criteria, One Construction

Both methods build new variables $z = X\phi$ by maximising something, subject to unit length and to not repeating earlier directions.

Method	**Maximises**	**Uses y?**
PCA (and PCR)	$\mathrm{Var}(X\phi)$	no
PLS	$\widehat{\mathrm{Cov}}(X\phi, y)^2$	yes

The construction is identical. The only difference is what counts as a good direction, and that depends entirely on what you are going to do with the result — look at the data, or predict something.

Where the Other Names Fit

All three methods so far assume the interesting structure is a straight line through the data. Not everything is.

1. Kernel PCA bend the space first, then run PCA there — handles curved structure such as concentric rings.
2. *t*-SNE place points in 2D so that each point's *near neighbours* stay near. Nothing else is preserved.
3. UMAP same ambition, different machinery; faster, and keeps a little more of the large-scale layout.
4. Autoencoders learn a nonlinear compression with a neural network. PCA is the case where it is linear.

Four Things to Take Away

1. A direction is a unit vector, and projecting onto it turns p numbers into one. PCA picks the direction whose projection keeps the most spread, because spread is what distinguishes observations.
2. The optimisation becomes an eigenproblem, and the eigenvalue *is* the variance along its eigenvector — which is why we take the largest.
3. PCA is not scale invariant. Variance has units; change the units and you change the answer. Standardise unless the variables are genuinely comparable.
4. PCA never sees y, so it can discard exactly the direction that mattered. PLS is the same construction with a criterion that weights each predictor by how strongly it moves with the response.